Mathematical Analysis Lecture 29 定积分的应用

三. 已知平行截面面积求体积. $V=\int_a^b S(x)\,dx \leftarrow dv=S(x)\,dx$.

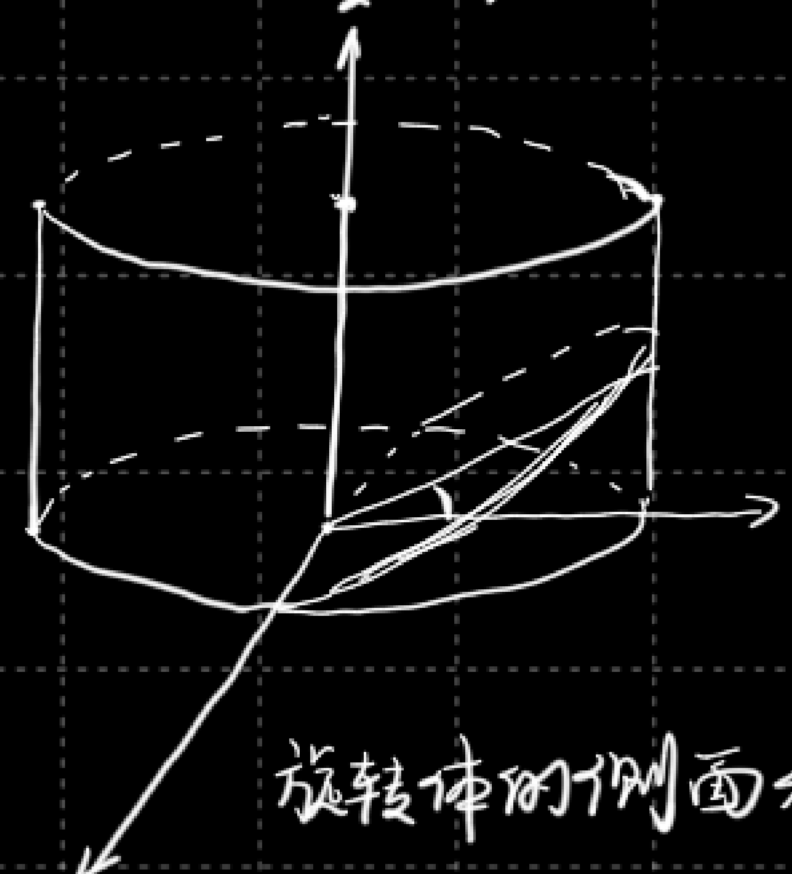
当V为由$y=f(x)$旋转而得, $V=\int_a^b \pi f^2(x)\,dx$.

绕y轴旋转,则: $x=f^{-1}(y)$ $V=\int_{f(a)}^{f(b)} \pi (f^{-1}(y))^2\,dx$.

e.g. 求 $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ 椭球体积.

1° $V=2\pi\int_0^a \left(b^2-\frac{b^2}{a^2}x^2\right)dx = 2\pi\left(b^2x-\frac{b^2}{3a^2}x^3\right)\Big|_0^a=\frac{4}{3}\pi ab^2$.

2° $x=a\cos x$, $y=b\sin x$ ∴ $S=\pi y^2=\pi b^2\sin^2 x$ ∴ $V=2\pi\int_0^{\frac{\pi}{2}} ab^2\sin^3 x\,dx = 2\pi\int_0^{\frac{\pi}{2}} ab^2(-\sin^2 x)\,d\cos x = \frac{4}{3}\pi ab^2$.



e.g. α角,求楔体积.

(自行用直角系求解).

旋转体的侧面积. 取$[x, x+dx]$上圆台的侧面积, $dS=2\pi y\,ds$, ds为小弧长: $ds=\sqrt{1+f'^2(x)}\,dx$.

∴ $dS=2\pi f(x)\sqrt{1+f'^2(x)}\,dx$ $S=\int_a^b 2\pi f(x)\sqrt{1+f'^2(x)}\,dx$.

若采用参方: $S=\int_{t_0}^{t_1}2\pi\psi(t)\sqrt{\varphi'^2(t)+\psi'^2(t)}\,dt$. $\begin{cases}x=\varphi(t)\\ y=\psi(t)\end{cases}$

在物理: 1° 变力沿直线所作的功

设物体在连续变力$F(x)$沿着x轴 $a\to b$, 力的方向与运动方向平行 求功

$W=\int_a^b F(x)\,dx$, $dW=F(x)\,dx$.

e.g. $\int_a^b \frac{kqe}{r^2}\,dr = -\frac{kqe}{r}\Big|_a^b = \frac{kqe}{a}-\frac{kqe}{b}$

$W=\int_0^5 9\pi\rho g\, x\,dx = \frac{9}{2}\pi\rho x^2 g\Big|_0^5 = \frac{225}{2}\pi\rho g$.